Aalok Thakkar

Department of Computer Science, Ashoka University, Sonipat, Haryana, India
thakkar@ashoka.edu.in | aalok-thakkar.github.io

Academic Appointment

2024–present	Assistant Professor, Department of Computer Science, Ashoka University, Sonipat.

Education

2018–2023	Ph.D., Computer and Information Science, University of Pennsylvania. Thesis: *Example-Guided Synthesis of Relational Queries.* Advisors: Rajeev Alur and Mayur Naik.
2015–2018	B.Sc. (Honours), Mathematics and Computer Science, Chennai Mathematical Institute.

Awards and Honours

2026	**Distinguished Paper**, Automated Technology for Verification and Analysis (ATVA 2026), for *How Often Does Your Program Fail?*
2026	**Best Paper Award**, Conference for AI Scientists (CAISc 2026), for *A Forced-Structure Reduction and Verifiable Bounds for Conway's 99-Graph.*
2026–2027	**INSA Visiting Scientist**, Indian National Science Academy. Residency at the Chennai Mathematical Institute to work with Aiswarya on the synthesis of string transducers.

Grants and Fellowships

2026–2029	**VSAC Young Faculty Grant**, Ashoka University.
2026	**VSAC Transdisciplinary Research Grant**, Ashoka University. Two projects: Circle of Joy (with Devapriya Roy and Annie Baxi) and Animal Acoustics (with Meghna Agarwala and Nandini Chatterjee Singh).
2026–	**ANRF Prime Minister Early Career Research Grant**, Anusandhan National Research Foundation. Robust and scalable program synthesis.
2026	**Nagarathna Memorial Grant**. Translation of *Nayi Kavita* poets into English.
2025–2026	**Data & AI Adoption Fellowship**, EkStep Foundation. AI safety benchmarks for education and agriculture.

2025–2026 **Darjeeling Revival Fellowship**, Alt Carbon.
Geostatistical modelling for enhanced rock weathering.

2025–2026 **Adaptive and Inclusive Pedagogy Grant**, Mphasis.
Curriculum design for an adaptive and inclusive classroom.

Publications

Conference Publications

Hardness of 'Infinite Energy' Puzzles. with Utkarsh Tiwari, Utkarsh Thanvi, Neeldhara Misra, and Armaan Sethi. *International Conference on Current Trends in Theory and Practice of Computer Science (SOFSEM 2027).*

Bhartrhari's Limitation Theorem. with Rhea. *Indian Conference on Logic and its Applications (ICLA 2027).*

The Trust a Protocol Needs. with Bhumika Mittal. *International Conference on Information Systems Security (ICISS 2026).*

Defining Purpose Limited Secrets. with Bhumika Mittal. *International Conference on Information Systems Security (ICISS 2026).*

Localising Stochasticity in Weighted Automata. with Smayan Agarwal. *Foundations of Software Technology and Theoretical Computer Science (FSTTCS 2026).*

How Often Does Your Program Fail?. with Arnab Ray. *Automated Technology for Verification and Analysis (ATVA 2026).* ***Distinguished Paper.***

A Forced-Structure Reduction and Verifiable Bounds for Conway's 99-Graph. *Conference for AI Scientists (CAISc 2026).* ***Best Paper Award.***

A Correctness-First Introduction to Computer Science. with Vaani Goenka. *Innovation and Technology in Computer Science Education (ITiCSE 2026).*

A Good Rubber Duck Does Not Quack. with Ayush Thonge. *Innovation and Technology in Computer Science Education (ITiCSE 2026).*

Accessibility Beyond Accommodations. with Vaanee Tripathi. *ACM India Compute Conference (COMPUTE 2025).*

Ancient Algorithms for a Modern Curriculum. *ACM India Compute Conference (COMPUTE 2025).*

BOOP: Write Right Code. with Vaani Goenka. *ACM India Compute Conference (COMPUTE 2025).*

SocraticAI: Transforming LLMs into Guided CS Tutors Through Scaffolded Interaction. with Karthik Sunil. *ACM India Compute Conference (COMPUTE 2025).*

From Transformers to Weighted Automata: Towards Verification of Large Language Models. with Smayan Agarwal, Aslah Ahmad Faizi, and Shobhit Singh. *From Data to Models and Back (DATAMOD 2025).*

Mobius: Synthesizing Relational Queries with Recursive and Invented Predicates. with Nathaniel Sands, George Petrou, Rajeev Alur, Mayur Naik, and Mukund Raghothaman. *Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA 2023).*

Relational Query Synthesis ⋈ Decision Tree Learning. with Aaditya Naik, Adam Stein, Rajeev Alur, and Mayur Naik. *International Conference on Very Large Data Bases (VLDB 2023).*

Example-Guided Synthesis of Relational Queries. with Aaditya Naik, Nathaniel Sands, Mukund Raghothaman, Mayur Naik, and Rajeev Alur. *Programming Language Design and Implementation (PLDI 2021).*

Additive Collatz Trajectories. with Mrunmay Jagadale. *International Conference of the Jangjeon Mathematical Society (ICJMS 2016).*

Journal Publications

Reopening businesses and risk of COVID-19 transmission. with Ashley O'Donoghue, Tenzin Dechen, Whitney Pavlova, Michael Boals, Garba Moussa, Manvi Madan, Frank J. DeFalco, and Jennifer P. Stevens. *npj Digital Medicine 4, 51 (2021).*

Concurrency in Boolean networks. with Thomas Chatain, Stefan Haar, Juraj Kolcak, and Loic Pauleve. *Natural Computing 19, 91–109 (2020).*

Infinitude of Primes Using Formal Languages. *The American Mathematical Monthly 125:8, 745–749 (2018).*

Workshop Publications

Complexity of Relational Query Synthesis. with Rajeev Alur and Mayur Naik. *11th Workshop on Synthesis (SYNT 2022).*

Modular Synthesis of Reactive Systems. with Kedar S. Namjoshi and Richard Trefler. *9th Workshop on Synthesis (SYNT 2020).*

Verification of Rewriting-based Query Optimizers. with Krishnamurthy Balaji and Piyush Gupta. *16th International Workshop on Termination (WST 2018).*

Teaching

Monsoon 2026	CS-2212-1: Data Structures and Algorithms.
Summer 2026	FC-0306: Quantitative Reasoning and Mathematical Thinking.
Spring 2026	CS-3330: Theory of Computation. CS-3610: Information Security.
Monsoon 2025	CS-1102: Introduction to Computer Science. CS-5310: Symbolic Logic and Applications. CS-6101: Computer Science Research Methods and Seminars.
Summer 2025	FC-0306: Quantitative Reasoning and Mathematical Thinking. CS-1102: Introduction to Computer Science.
Spring 2025	CS-1102: Introduction to Computer Science. CS-5440: Trustworthy AI.
Monsoon 2024	CS-5310: Symbolic Logic and Applications.

Professional Service

Organiser	Verification, Logic, and Cryptography (VeLCro) 2026, CMI. ACM COMPUTE 2026, Amrita Vishwa Vidyapeetham. FSTTCS 2026, IIT Delhi. Formal Methods Update Meeting 2026, Krea University. ACM COMPUTE Regional 2026 (North India), Ashoka University. IndiCS Seminar 2: Automated Synthesis, 2025, Infosys Mysore (coordinator). FSTTCS 2024, FSTTCS 2025. IARCS Verification Seminar Series. Logic Society of Delhi.
Editor	Corresponding editor, *Bhāvanā* magazine (2026).
Reviewer	Programme committees and review activities for ICLA 2026, MFCS 2026, OOPSLA 2026, ATVA AEC 2025, PLDI SRC 2025, ISEC 2025, POPL AEC 2022, *ACM Computing Surveys*, and the *Indian Journal of History of Science*.
Memberships	Association for Computing Machinery (ACM India). Vijnana Bharati.